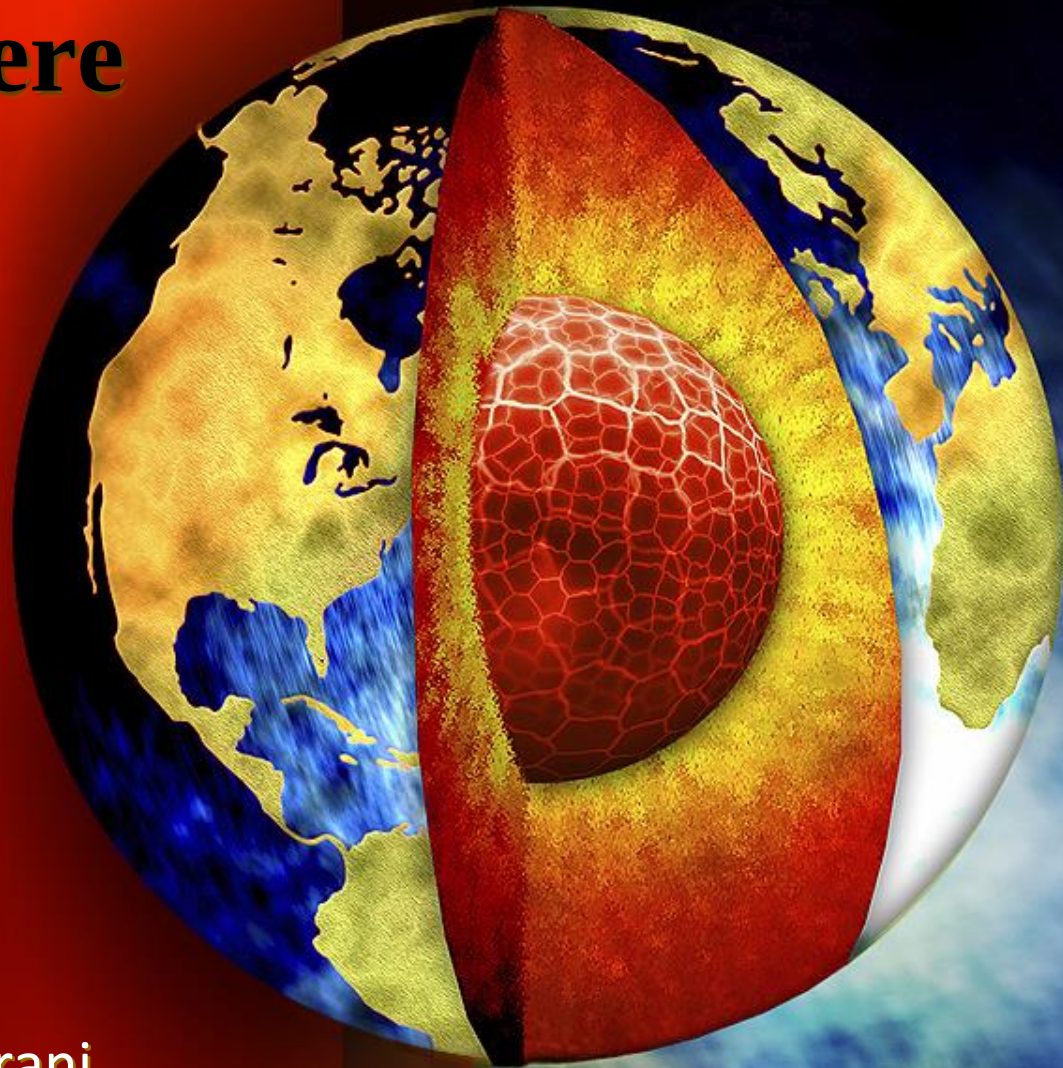
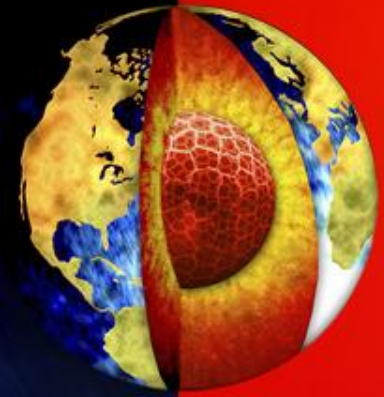
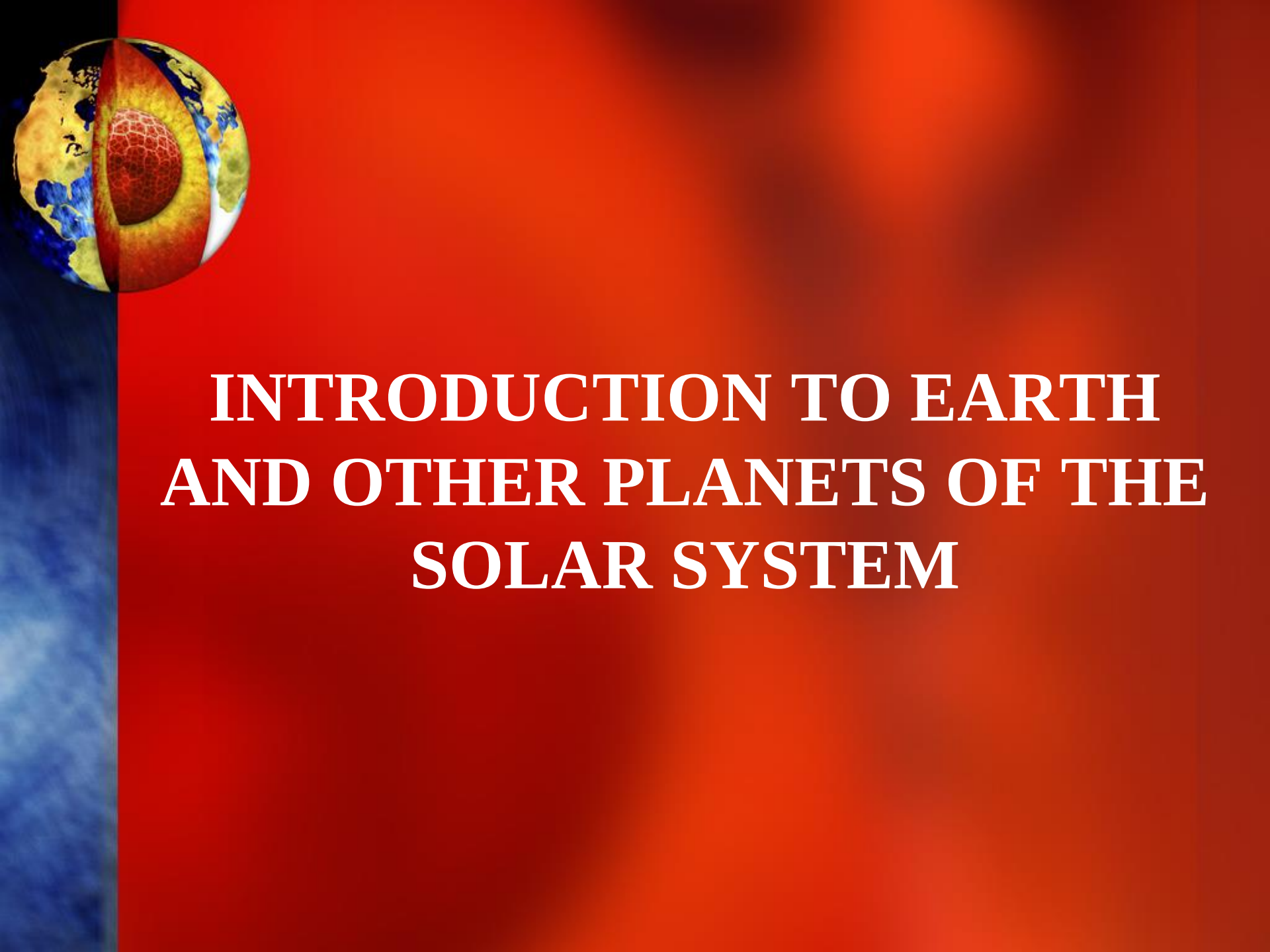


Lithosphere



Prabodha Punsarani

M.Phil in Geography (Reading), BA (Hons) in Geography, LLB (Hons)

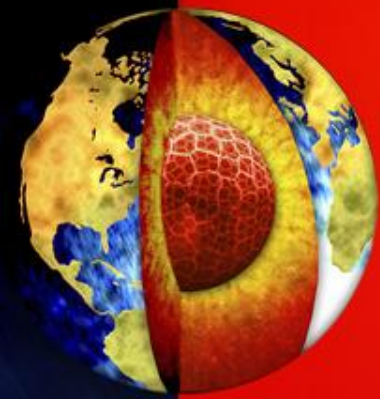


INTRODUCTION TO EARTH AND OTHER PLANETS OF THE SOLAR SYSTEM

The Definition of a Planet

- The word goes back to the ancient Greek word *planēt*, and it means "wanderer." (<https://science.nasa.gov>)
- A more modern definition can be found in the Merriam-Webster dictionary which defines a planet as "any of the large bodies that revolve around the Sun in the solar system." (<https://science.nasa.gov>)
- In 2006, the International Astronomical Union (IAU) - a group of astronomers that names objects in our solar system - agreed on their own definition of the word "planet."

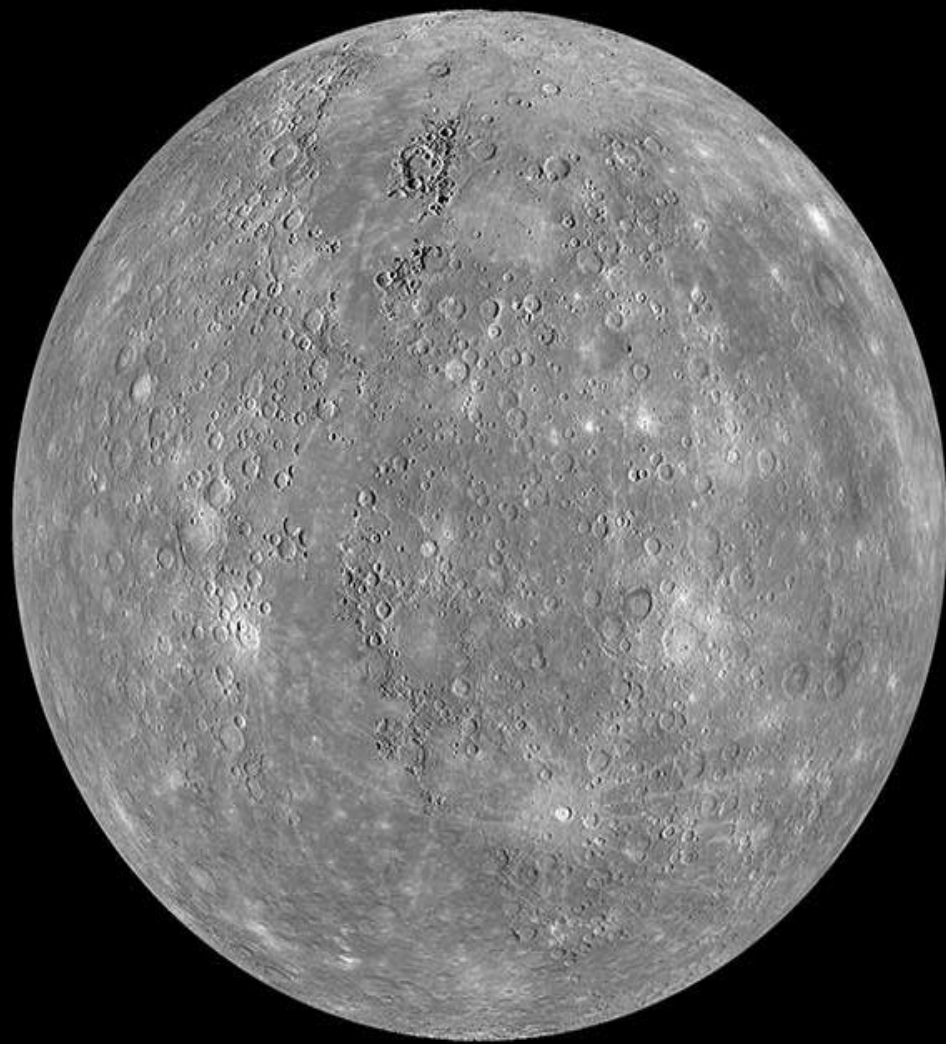




This new definition changed caused Pluto's famous "demotion" to a dwarf planet(<https://science.nasa.gov>)

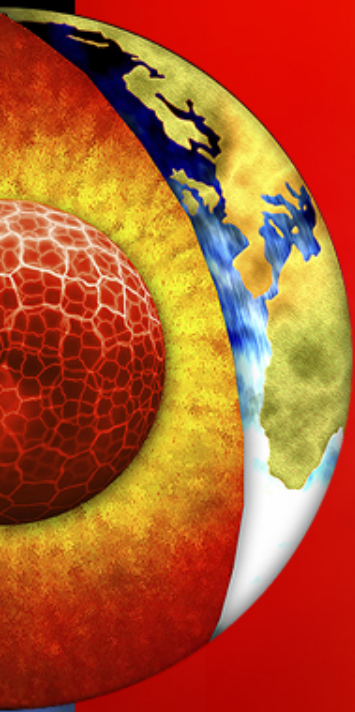
The definition of a planet adopted by the IAU says a planet must do three things:

- 1.It must orbit a star (in our cosmic neighborhood, the Sun).(<https://science.nasa.gov>)
- 2.It must be big enough to have enough gravity to force it into a spherical shape.
(<https://science.nasa.gov>)
- 3.It must be big enough that its gravity has cleared away any other objects of a similar size near its orbit around the Sun. (<https://science.nasa.gov>)



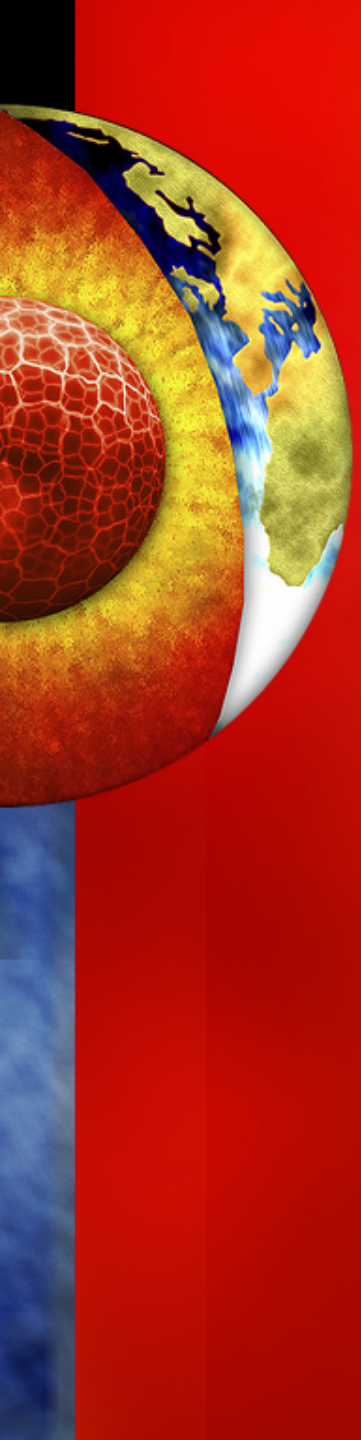
Mercury

Mercury

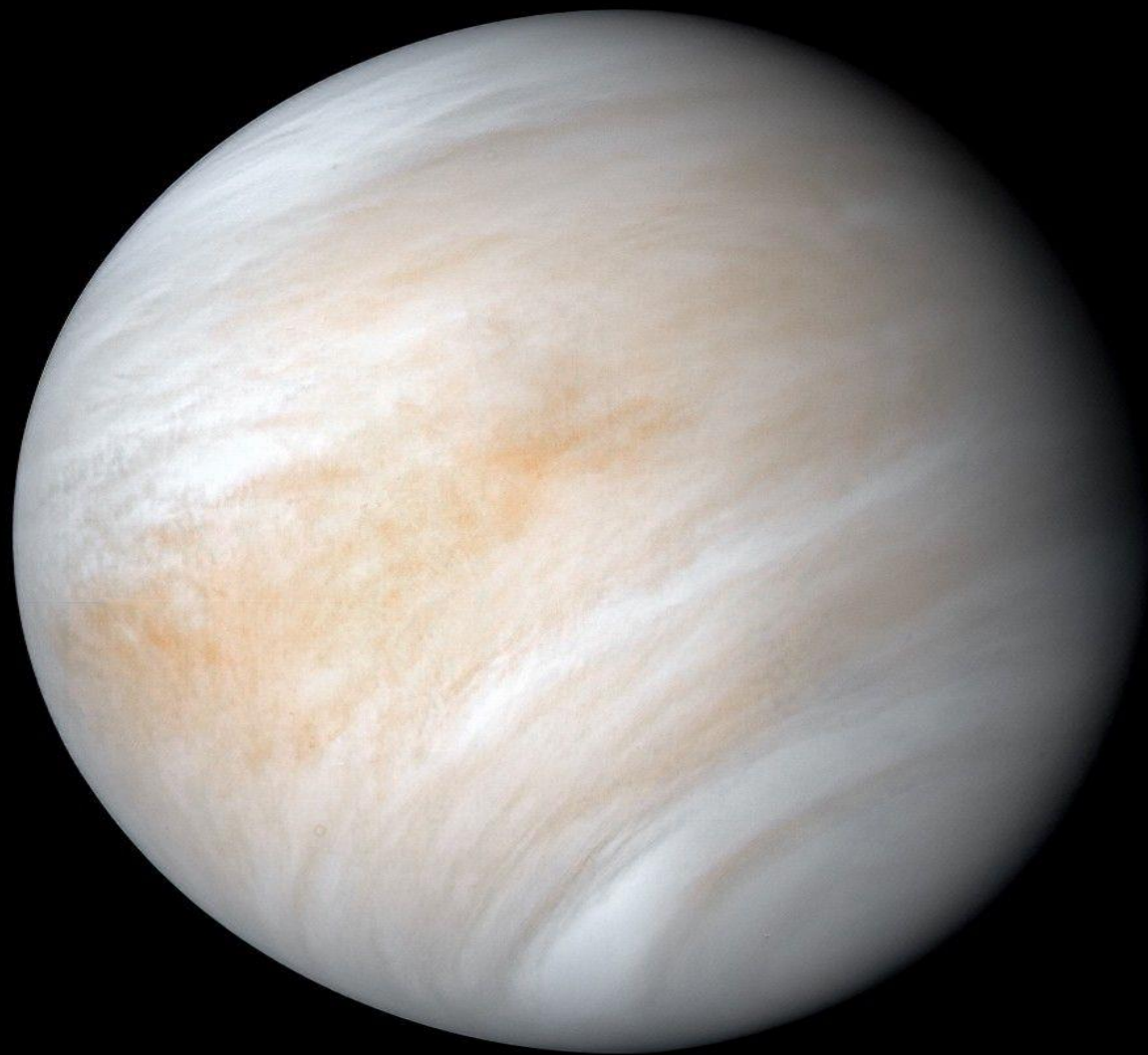
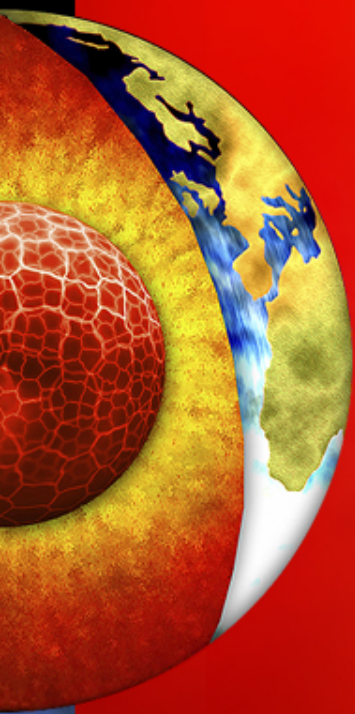


Mercury is the smallest planet in our solar system and nearest to the Sun. It's only slightly larger than Earth's Moon. From the surface of Mercury, the Sun would appear more than three times as large as it does when viewed from Earth, and the sunlight would be as much as seven times brighter.

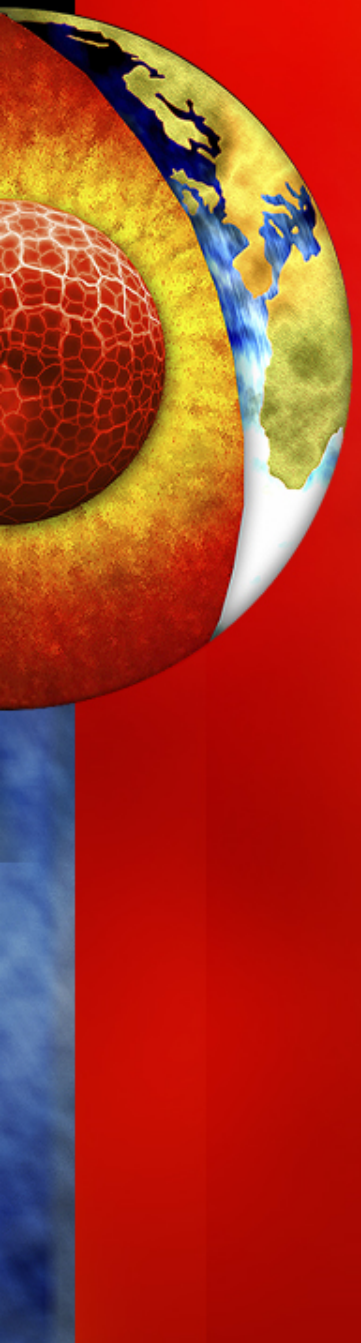
Mercury's surface temperatures are both extremely hot and cold.



- Because the planet is so close to the Sun, day temperatures can reach highs of 800°F (430°C). Without an atmosphere to retain that heat at night, temperatures can dip as low as -290°F (-180°C).
- Despite its proximity to the Sun, Mercury is not the hottest planet in our solar system – that title belongs to nearby Venus, thanks to its dense atmosphere. But Mercury is the fastest planet, zipping around the Sun every 88 Earth days.



Venus



Venus

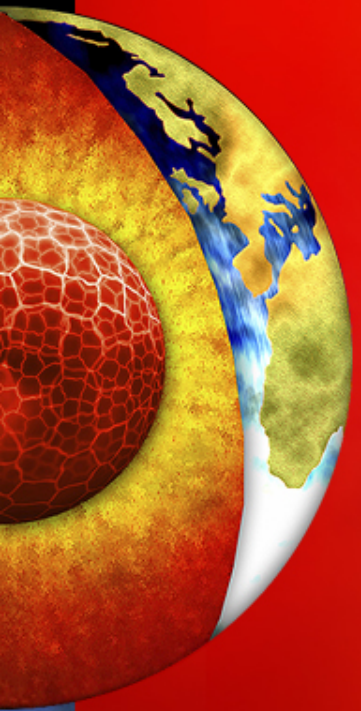
- Venus is the second planet from the Sun, and Earth's closest planetary neighbor. Venus is the third brightest object in the sky after the Sun and Moon. Venus spins slowly in the opposite direction from most planets.
- Venus is similar in structure and size to Earth, and is sometimes called Earth's evil twin. Its thick atmosphere traps heat in a runaway greenhouse effect, making it the hottest planet in our solar system with surface temperatures hot enough to melt lead. Below the dense, persistent clouds, the surface has volcanoes and deformed mountains.



<https://science.nasa.gov/earth/facts/>

Earth

Earth



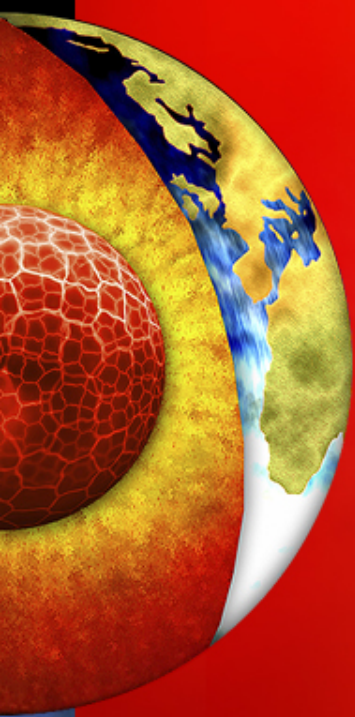
- While Earth is only the fifth largest planet in the solar system, it is the only world in our solar system with liquid water on the surface. Just slightly larger than nearby Venus, Earth is the biggest of the four planets closest to the Sun, all of which are made of rock and metal.
- Earth is the only planet in the solar system whose English name does not come from Greek or Roman mythology. The name was taken from Old English and Germanic. It simply means "the ground." There are, of course, many names for our planet in the thousands of languages spoken by the people of the third planet from the Sun.



<https://science.nasa.gov/mars/facts/>

Mars

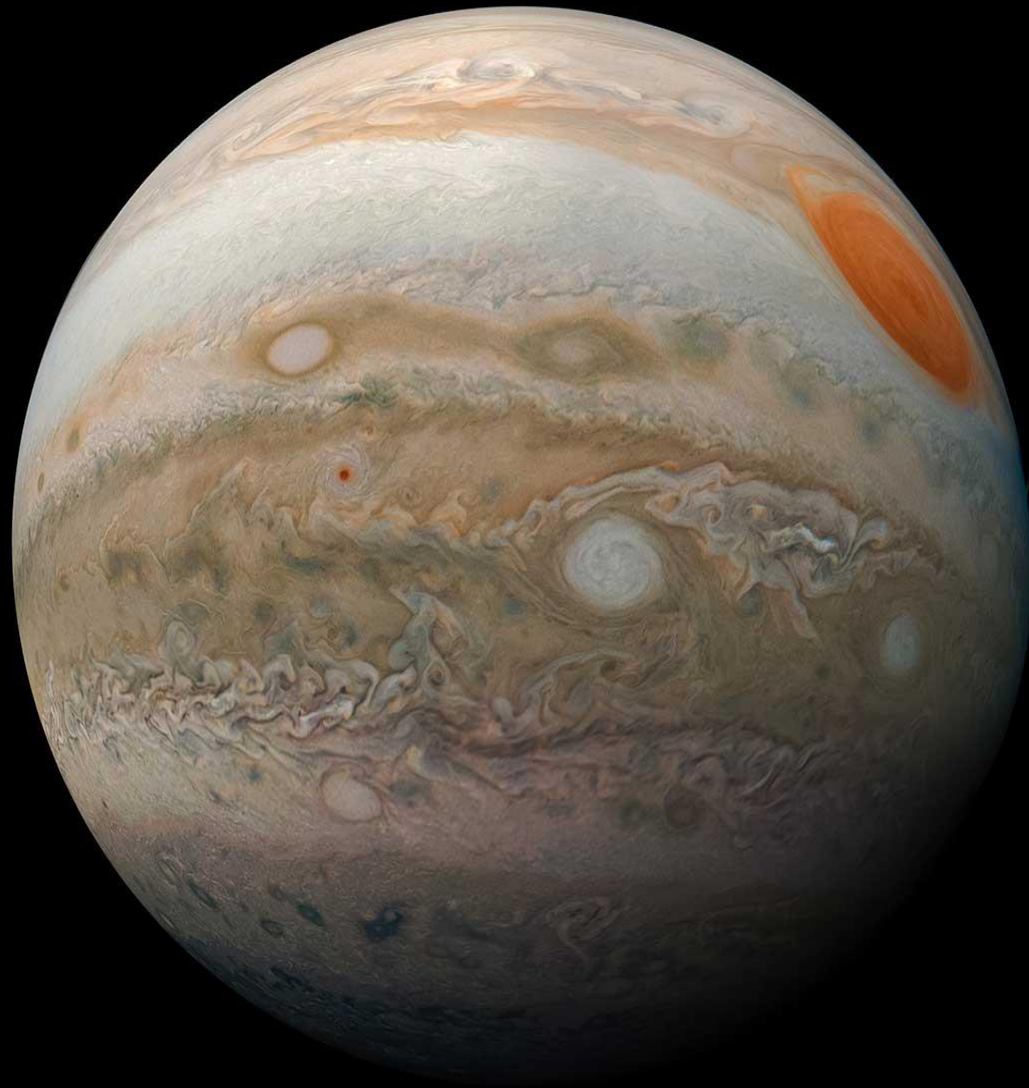
Mars



Mars is one of the most explored bodies in our solar system, and it's the only planet where we've sent rovers to roam the alien landscape. NASA missions have found lots of evidence that Mars was much wetter and warmer, with a thicker atmosphere, billions of years ago.

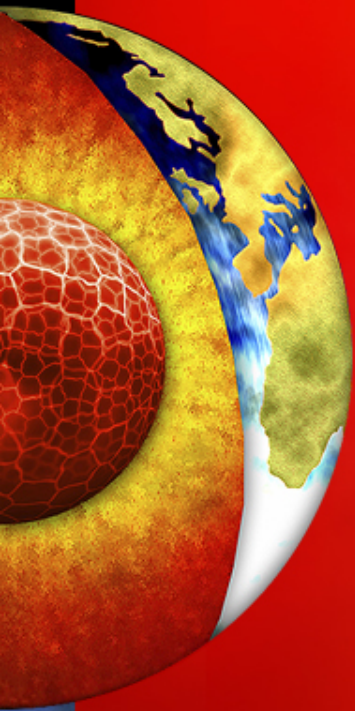
Mars was named by the Romans for their god of war because its reddish color was reminiscent of blood. The Egyptians called it "Her Desher," meaning "the red one."

Even today, it is frequently called the "Red Planet" because iron minerals in the Martian dirt oxidize, or rust, causing the surface to look red.



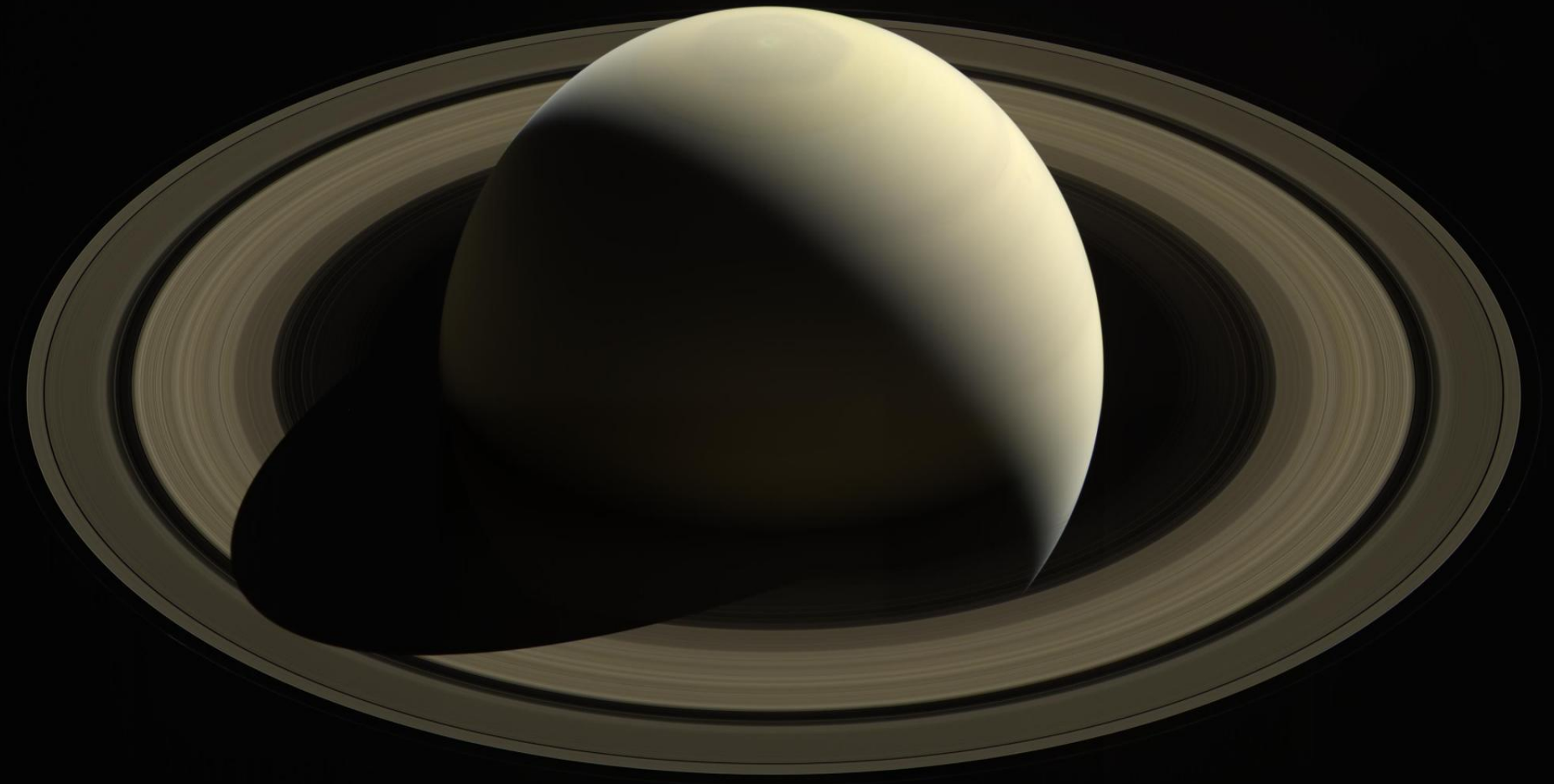
Jupiter

Jupiter



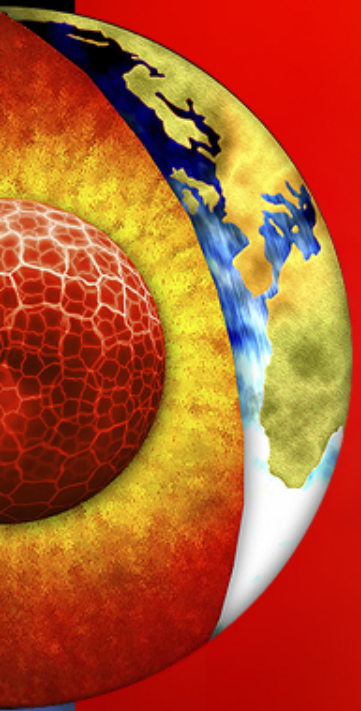
Jupiter is a world of extremes. It's the largest planet in our solar system – if it were a hollow shell, 1,000 Earths could fit inside. It's also the oldest planet, forming from the dust and gases left over from the Sun's formation 4.6 billion years ago. But it has the shortest day in the solar system, taking about 9.9 hours to spin around once on its axis.

Jupiter's signature stripes and swirls are actually cold, windy clouds of ammonia and water, floating in an atmosphere of hydrogen and helium. The dark orange stripes are called belts, while the lighter bands are called zones, and they flow east and west in opposite directions. Jupiter's iconic Great Red Spot is a giant storm bigger than Earth that has raged for hundreds of years.



Saturn

Saturn

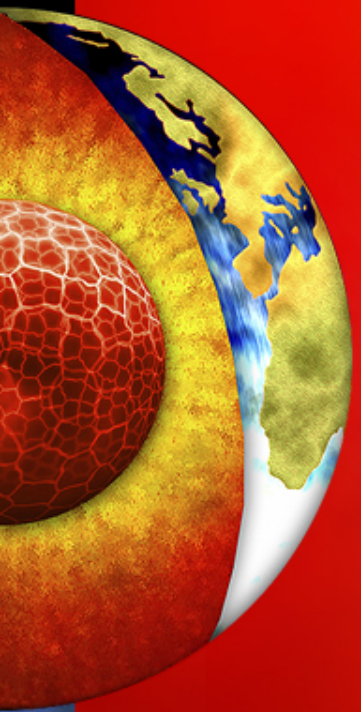


- Like fellow gas giant Jupiter, Saturn is a massive ball made mostly of hydrogen and helium. Saturn is not the only planet to have rings, but none are as spectacular or as complex as Saturn's. Saturn also has dozens of moons.
- From the jets of water that spray from Saturn's moon Enceladus to the methane lakes on smoggy Titan, the Saturn system is a rich source of scientific discovery and still holds many mysteries.

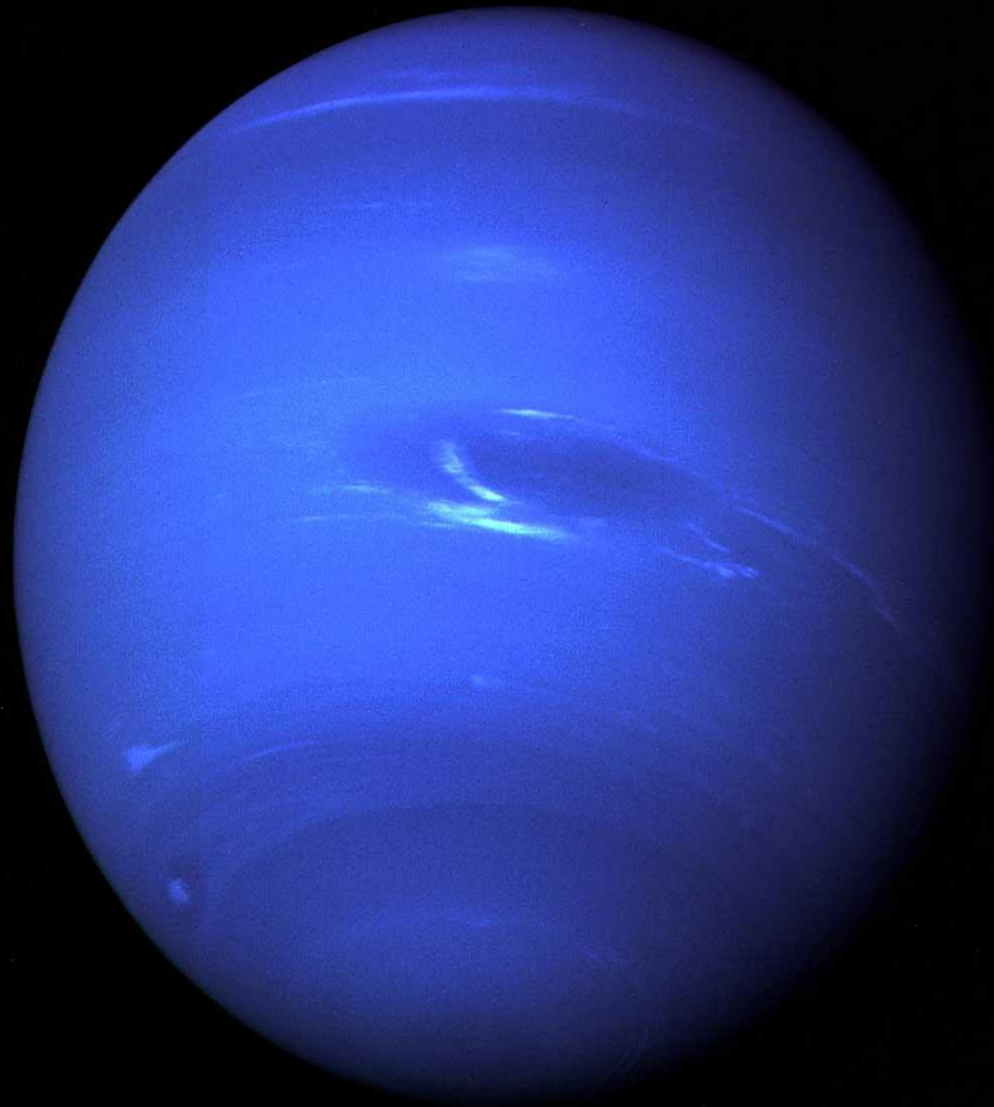


Uranus

Uranus

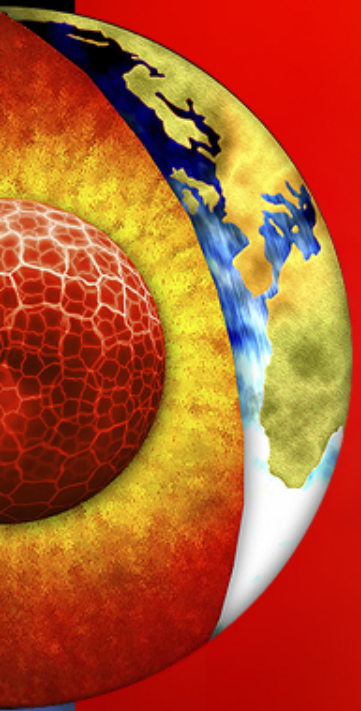


- Uranus is a very cold and windy world. The ice giant is surrounded by 13 faint rings and 28 small moons. Uranus rotates at a nearly 90-degree angle from the plane of its orbit. This unique tilt makes Uranus appear to spin sideways, orbiting the Sun like a rolling ball.
- Uranus was the first planet found with the aid of a telescope. It was discovered in 1781 by astronomer William Herschel, although he originally thought it was either a comet or a star. It was two years later that the object was universally accepted as a new planet, in part because of observations by astronomer Johann Elert Bode.

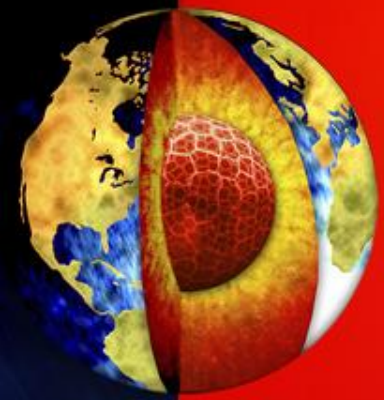


Neptune

Neptune



- Dark, cold, and whipped by supersonic winds, ice giant Neptune is more than 30 times as far from the Sun as Earth. Neptune is the only planet in our solar system not visible to the naked eye. In 2011 Neptune completed its first 165-year orbit since its discovery in 1846.
- Neptune is so far from the Sun that high noon on the big blue planet would seem like dim twilight to us. The warm light we see here on our home planet is roughly 900 times as bright as sunlight on Neptune.



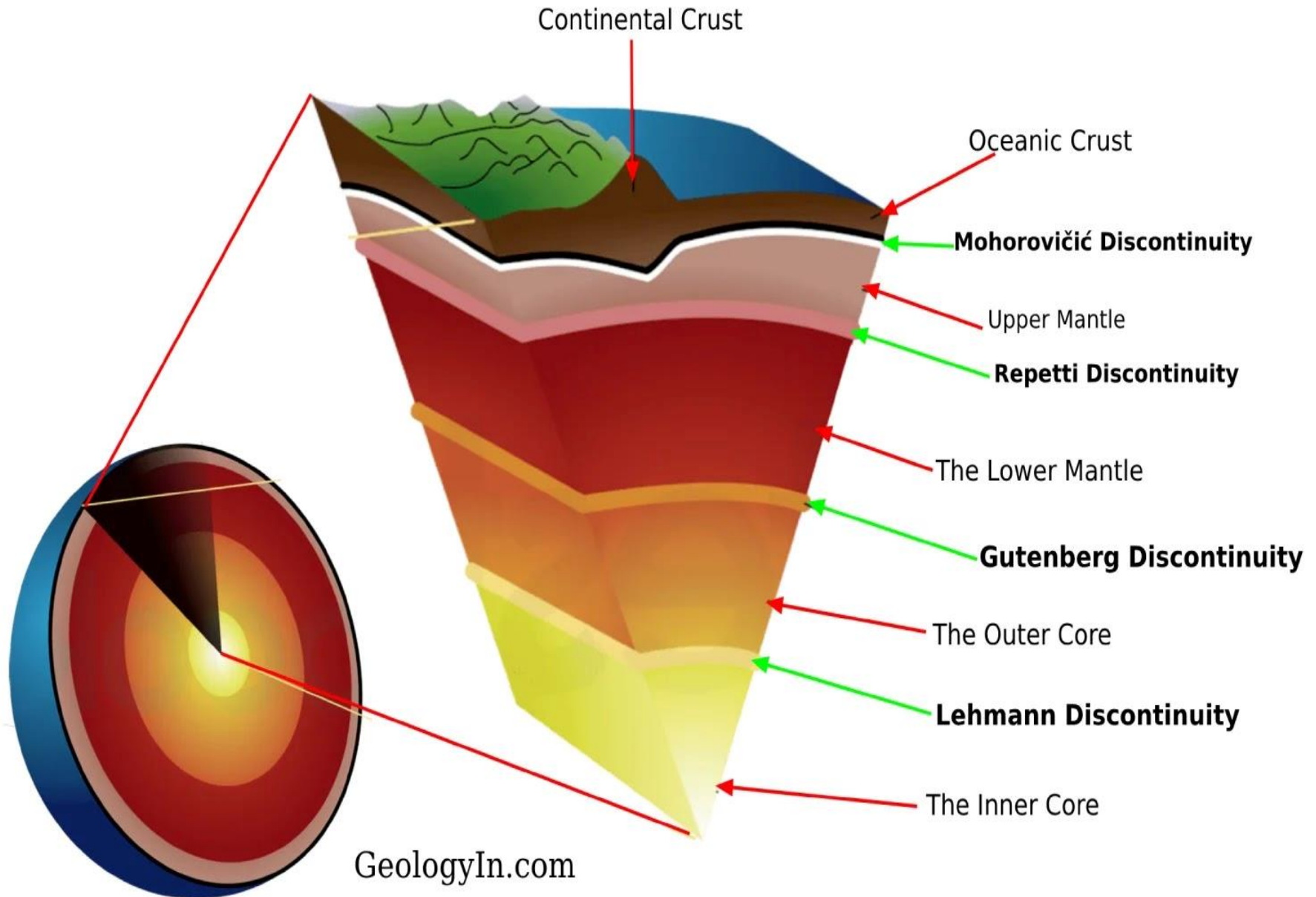
How Cutting Different
Planets is like 🤨

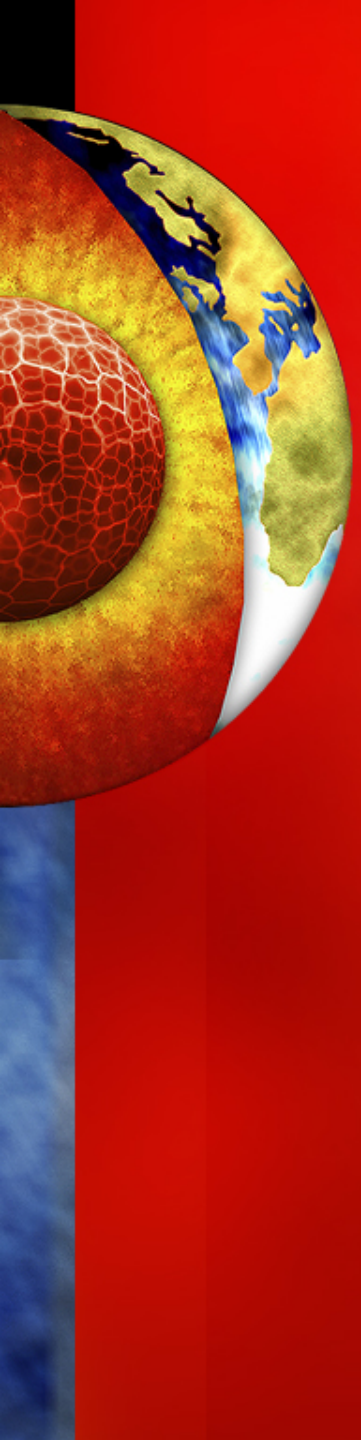




NATURE AND PROCESSES OF EARTH STRUCTURE

Discontinuities Inside the Earth





Inner Core

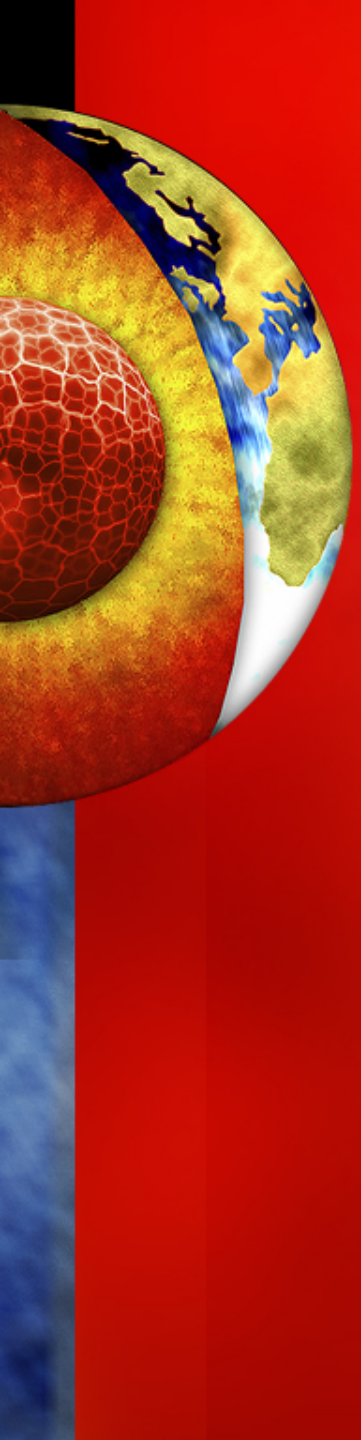
Solid iron & nickel. Extends from ~5,150 km to Earth's center (~6,371 km).

Lehmann Discontinuity

- At ~5,150 km depth.
- Boundary between **outer (liquid) core** and **inner (solid) core**.
- P-wave speeds increase again.

Outer Core

- Liquid iron & nickel.
- Responsible for Earth's magnetic field.
- Extends from ~2,900 km to ~5,150 km.



Gutenberg Discontinuity

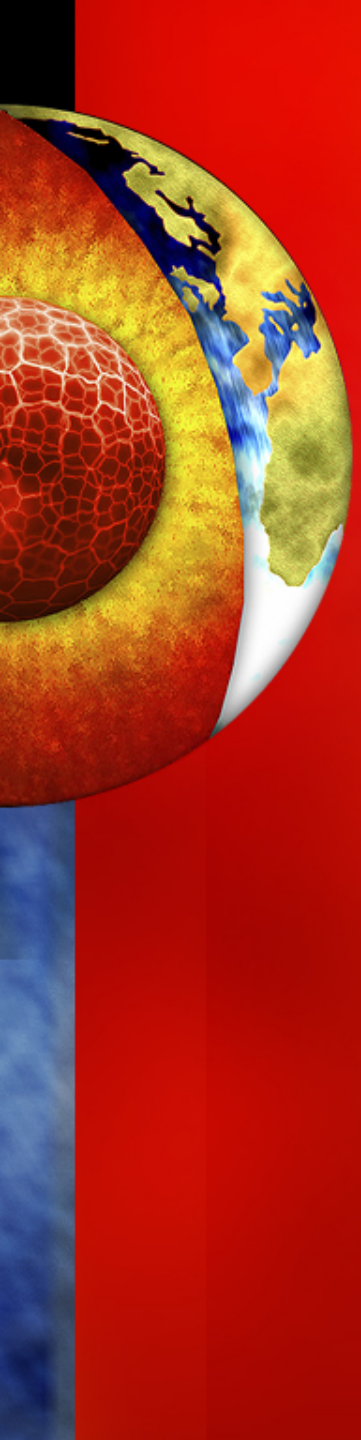
- At ~2,900 km depth.
- Boundary between mantle and outer core.
- Seismic S-waves stop (since the outer core is liquid), P-waves slow down.

Lower Mantle

- Extends from 660 km to ~2,900 km.
- More rigid than the asthenosphere but still very slow-flowing.

Upper Mantle

Solid but can flow very slowly (plastic). Made of peridotite.

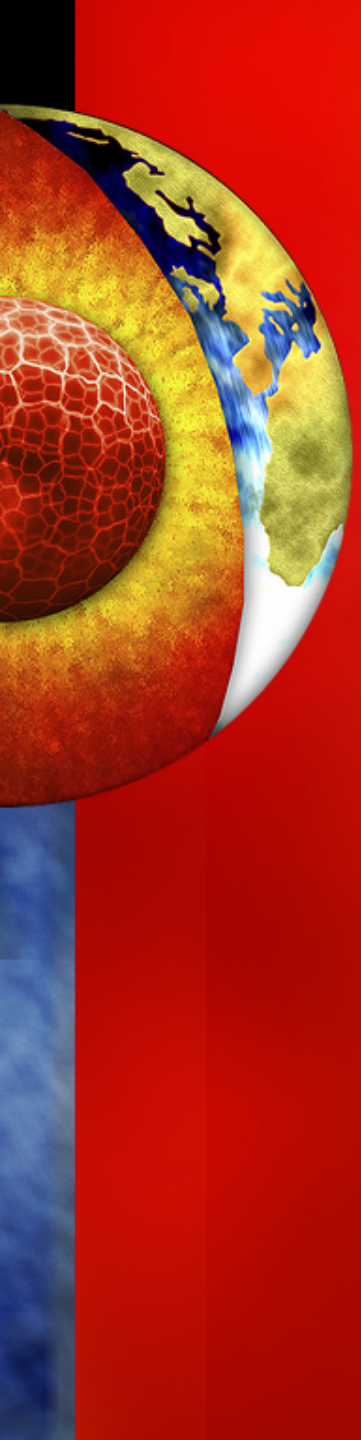


Mohorovičić (Moho) Discontinuity

- Boundary between **crust** and **mantle**.
- Seismic waves suddenly increase speed here because mantle rocks are denser.
- Depth: ~35 km under continents; ~7 km under oceans.

Crust

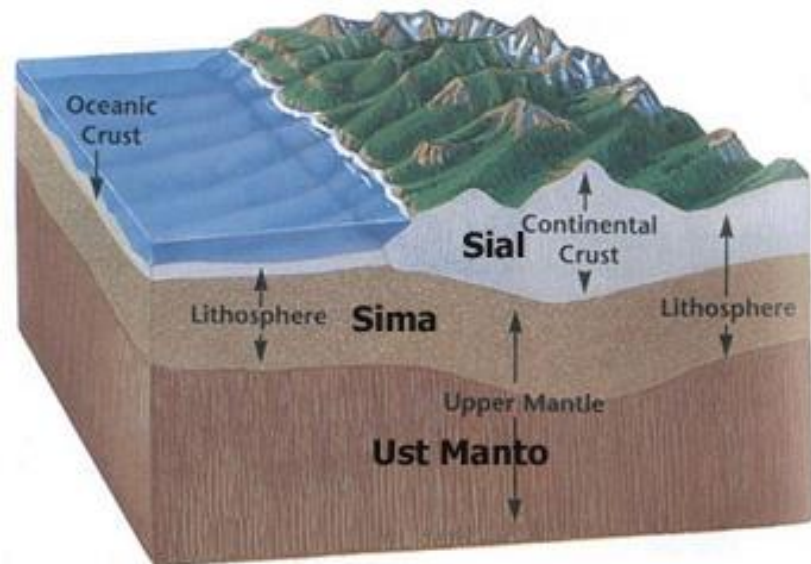
Outermost, solid layer. Made of rocks like granite (continental crust) and basalt (oceanic crust). Thickness: Oceanic: ~5–10 km
Continental: ~30–50 km

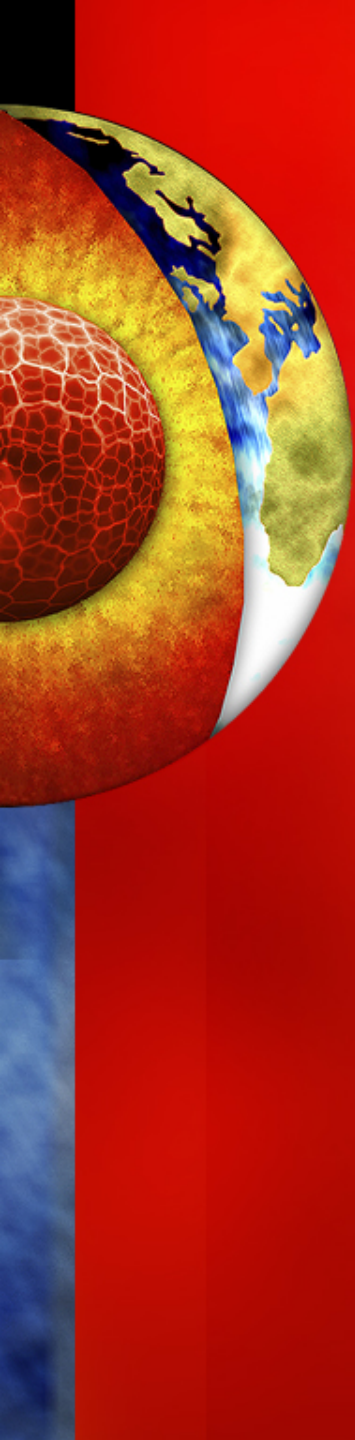


Earth's crust is divided into two types: oceanic crust and continental crust.

The transition zone between these two types of crust is sometimes called the Conrad discontinuity.

Silicates (mostly compounds made of silicon and oxygen) are the most abundant rocks and minerals in both oceanic and continental crust.





Oceanic crust

Oceanic crust, extending 5 to ten kilometers (3-6 kilometers) beneath the ocean floor, is mostly composed of different types of basalts. Geologists often refer to the rocks of the oceanic crust as “sima.” Sima stands for silicate and magnesium, the most abundant minerals in oceanic crust. (Basalts are a sima rocks.)

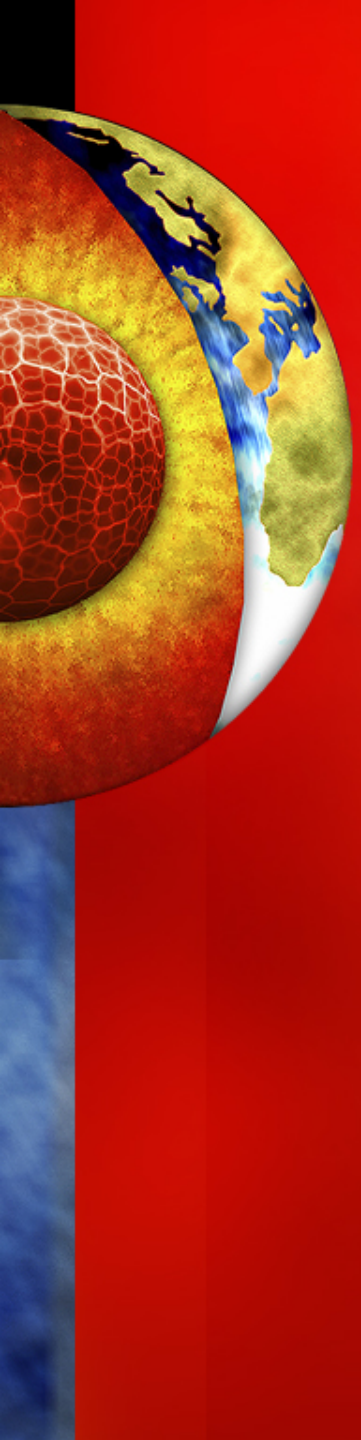
Its age and density increase with distance from ridges. Oceanic crust is much younger than continental crust; the oldest oceanic crust (~270 million years)

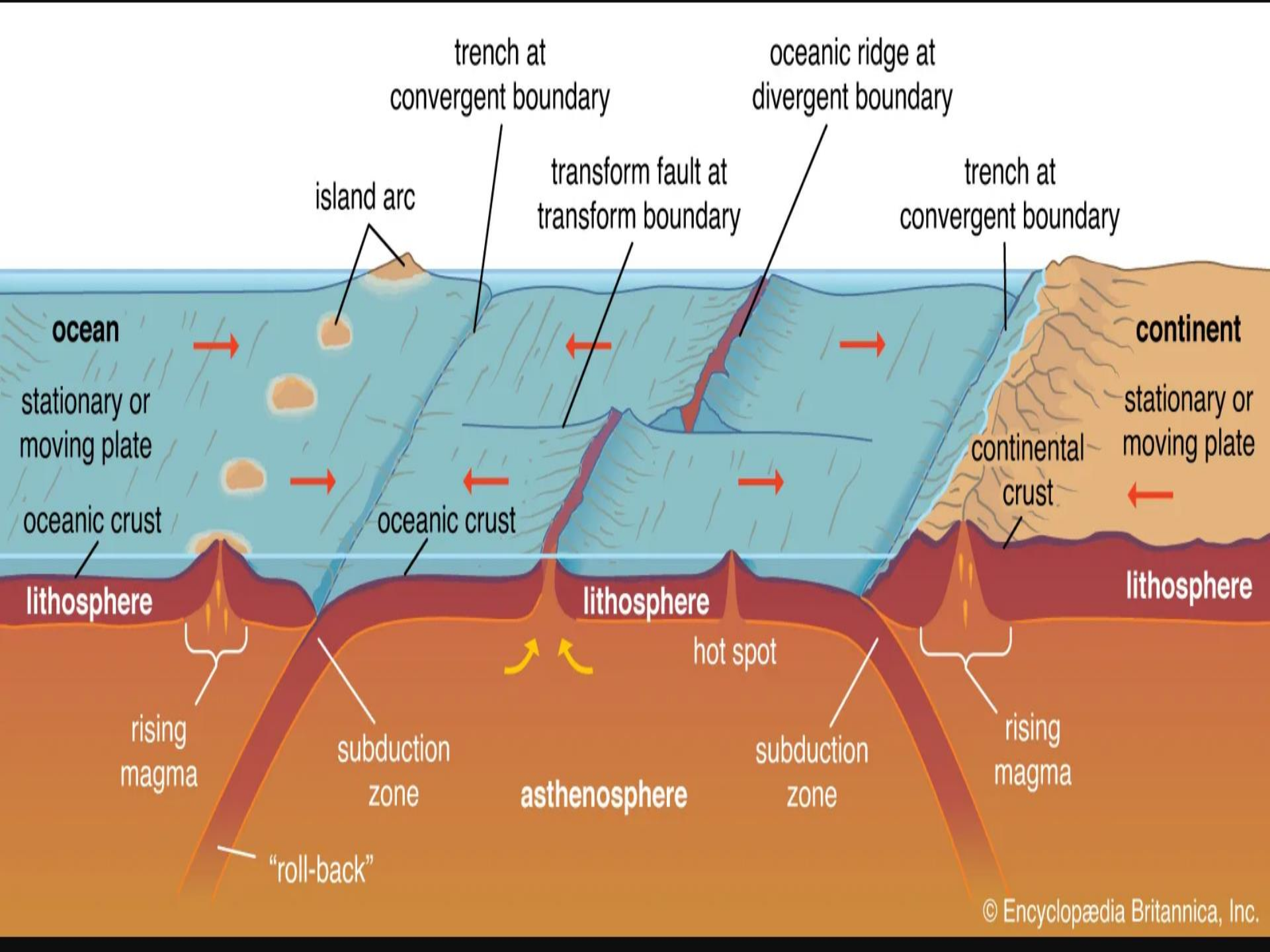
Geologists study oceanic crust by drilling, using submersibles, and examining ophiolites — sections of oceanic crust uplifted onto land.

Continental Crust

Continental crust is mostly composed of different types of granites. Geologists often refer to the rocks of the continental crust as “sial.” Sial stands for silicate and aluminum, the most abundant minerals in continental crust. Sial can be much thicker than sima.

Continental crust is formed at convergent boundaries, where plates collide and create mountains (orogeny). The thickest continental crust is beneath tall mountain ranges like the Himalayas and Andes, extending deep below the surface. Continental crust is much older than oceanic crust, with some parts nearly as old as Earth.







CONTINENTAL DRIFT THEORY BY ALFRED WEGENER



225 million years ago



150 million years ago



100 million years ago



Earth today

© 2007 EB Inc.

Continental Drift



Mountains



Land



Shallow seas



Ocean basins



Subduction
zones

legend



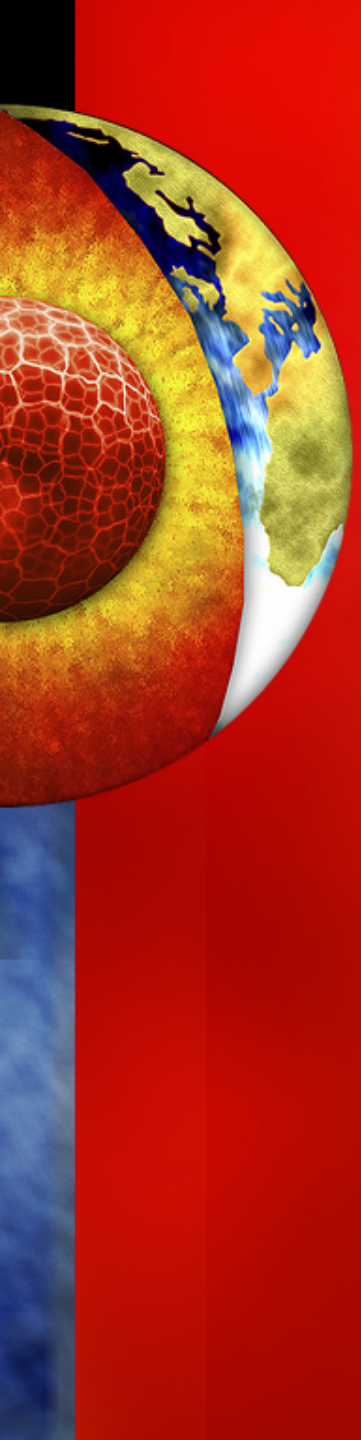
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- Continental drift describes one of the earliest ways geologists thought continents moved over time. Today, the theory of continental drift has been replaced by the science of plate tectonics.

The theory of continental drift is most associated with the scientist Alfred Wegener. In the early 20th century, Wegener published a paper explaining his theory that the continental landmasses were “drifting” across the Earth, sometimes plowing through oceans and into each other. He called this movement continental drift.

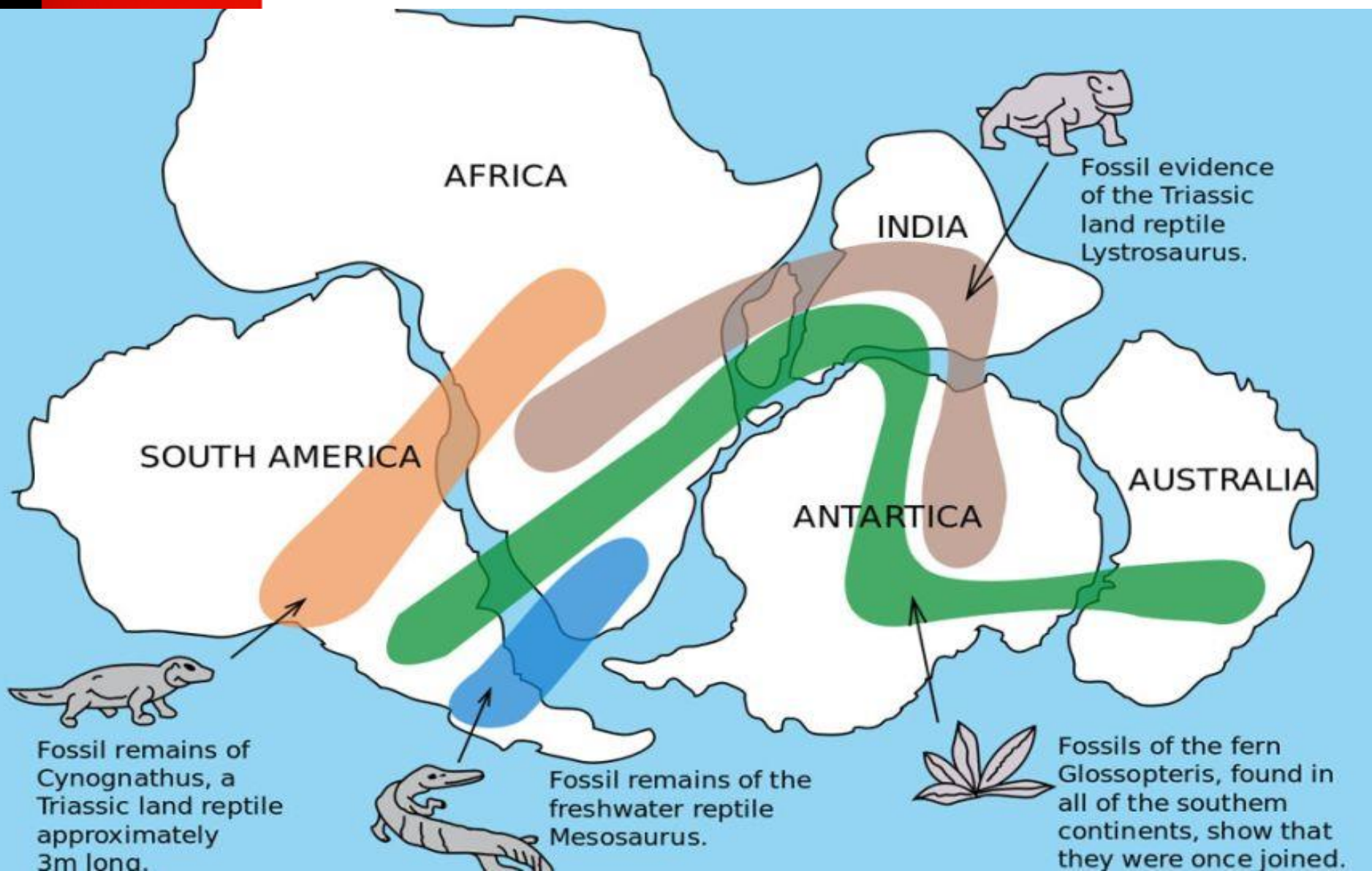


Pangaea:

- Wegener proposed that approximately 200 million years ago, all of Earth's continents were once part of an enormous, single supercontinent landmass called **Pangaea**.

Panthalassa:

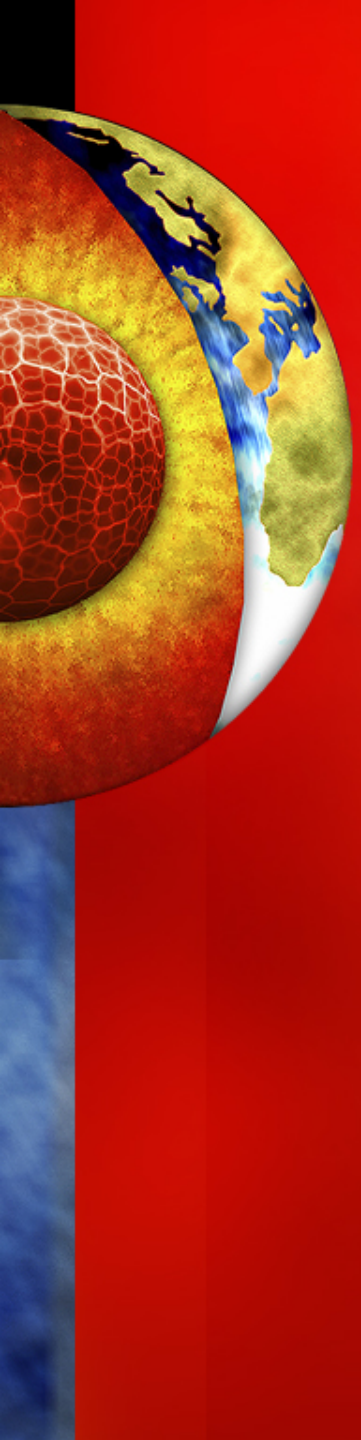
- The supercontinent **Pangaea** was surrounded by the mega-ocean **Panthalassa**, shaping Earth's ancient geography.
- He argued that, around 200 million years ago, Pangaea split into two large continental masses **Laurasia** and **Gondwanaland** forming the **northern** and **southern** components.
- Laurasia and Gondwanaland further evolved into present-day continents.

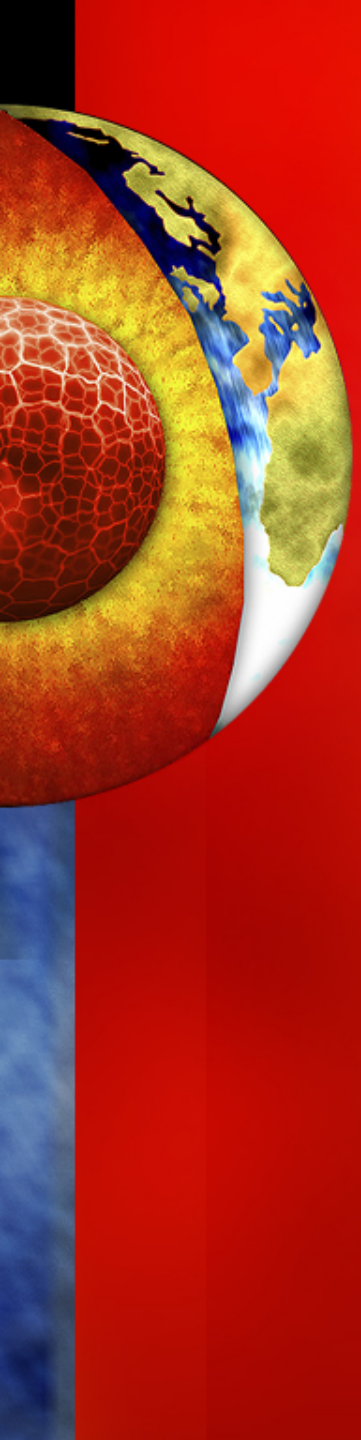


Supporting Evidence for Continental Drift

Jig-Saw Puzzle: Matching shorelines of Africa and South America.

- Wegener observed that the rock layers and mountain ranges on the east coast of South America and the west coast of Africa fit together, similar to a jigsaw puzzle.
- He also found geological similarities between the Appalachian Mountains in the eastern United States and the Caledonian Mountains in Scotland.





Rocks of the Same Age across the Oceans:

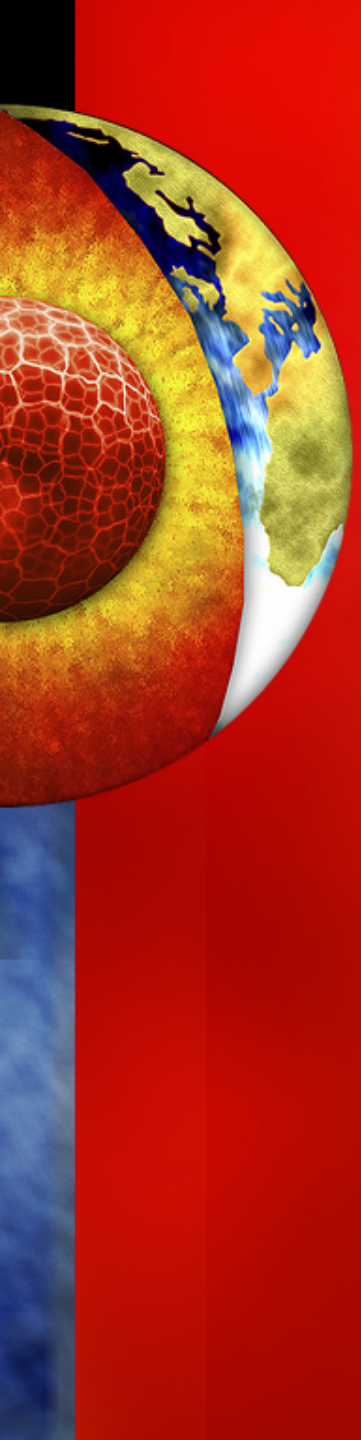
A 2,000-million-year-old rock belt from Brazil has similar formations as in western Africa.

Earliest Marine Deposits:

It was found along **South America** and **Africa's coastlines**, from the **Jurassic age**, indicate the absence of an ocean before then.

Placer Deposits:

The **Ghanaian coast** has rich gold placer deposits despite the **absence of a gold source** rock. The **Brazilian plateau** has **gold bearing veins** from which they are derived.



Tillite:

These are **Sedimentary rocks** formed out of **glacier deposits**.

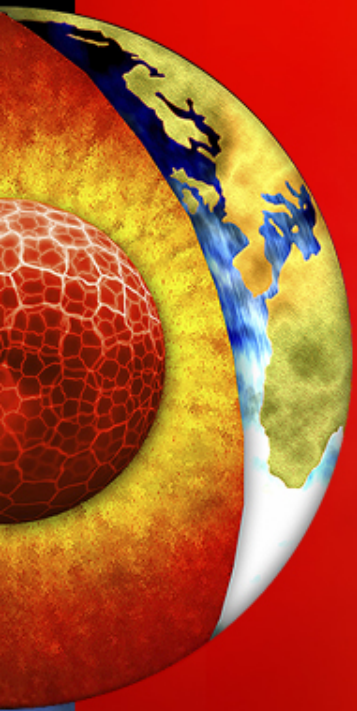
Notably, **Gondawana sediments** with tillite at the base from India have counterparts in regions like Africa, Madagascar, and Antarctica, implying shared geological histories.

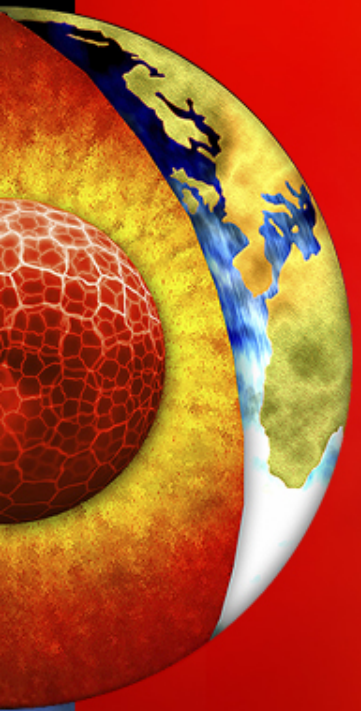
- These Glacial tillites offer clear evidence of **past climates** and **continental drift**.

Distribution of Fossils: Lemurs,

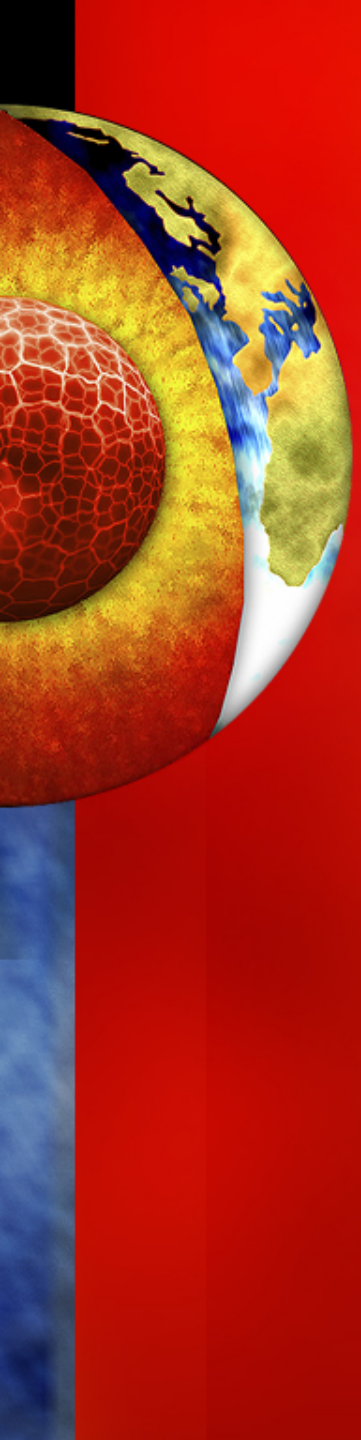
for instance, are found across India, Madagascar, and Africa even though these places are now separated by ocean.

- Mesosaurus, (a small reptile) is found only in two places: **Southern Cape province** of South Africa and **Iraver formations in Brazil** which are now 4,800 km apart with an ocean between them.
- Wegener studied **tropical plant fossils in Norway's Svalbard**, indicating a warmer past climate for the Arctic region.

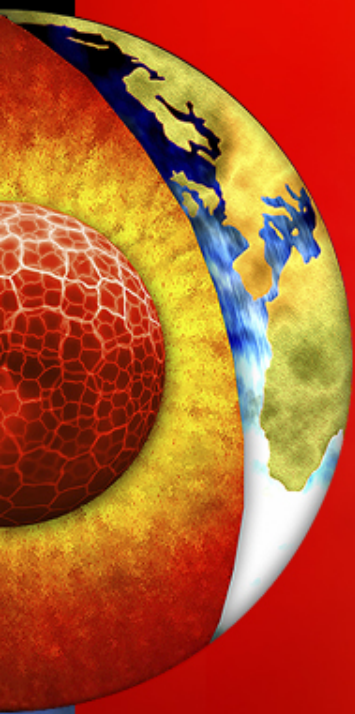




Why was the continental drift theory rejected?



THEORY OF PLATE TECTONICS



How Earth will look in 250 million Years

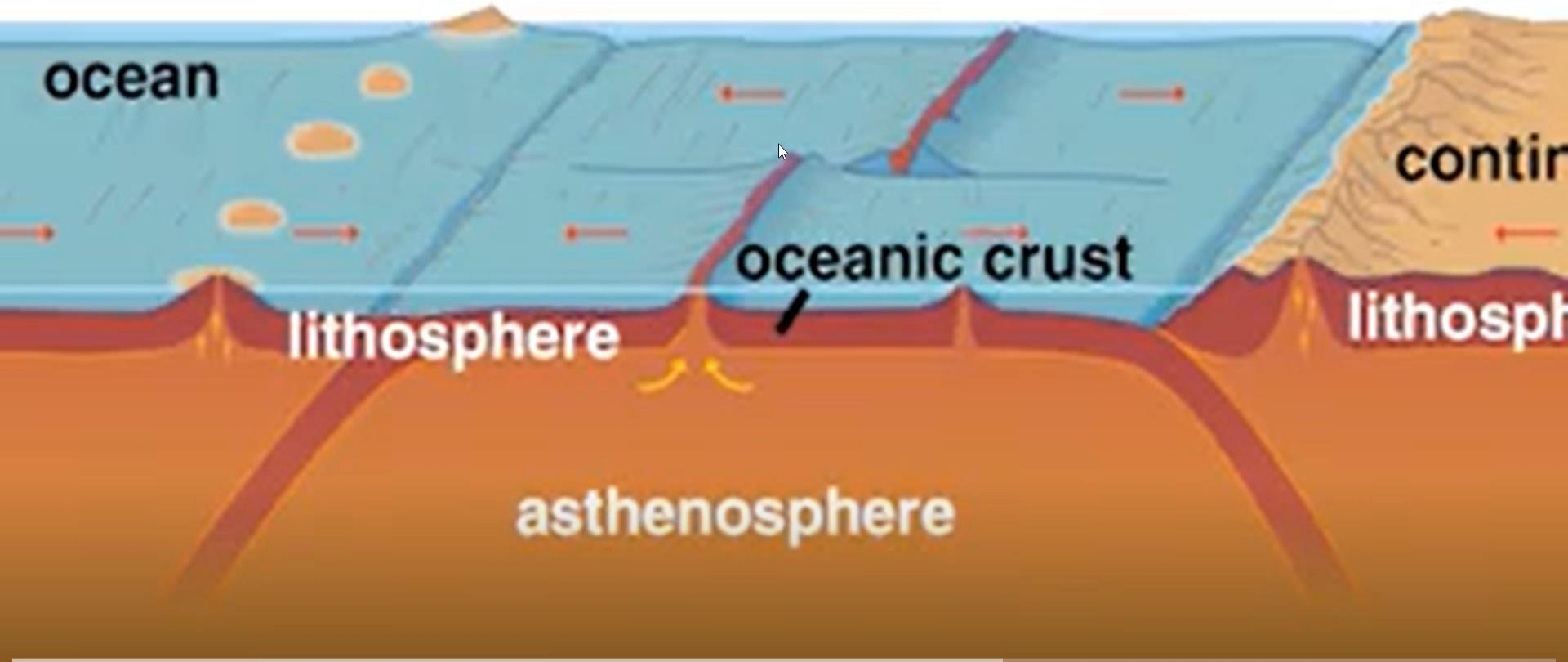


TECH
INSIDER

@SpaceScience @CosmicVisa

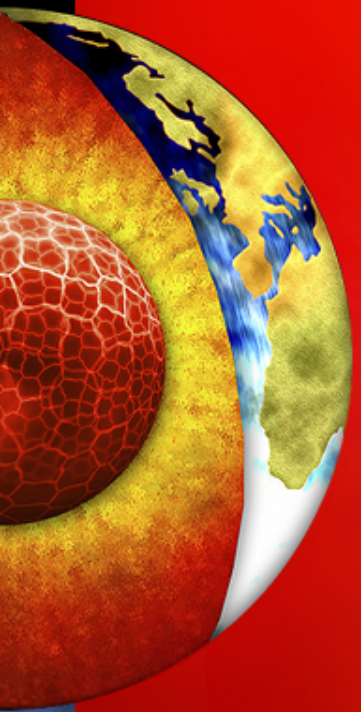


AGU
ADVANCING EARTH AND SPACE SCIENCE

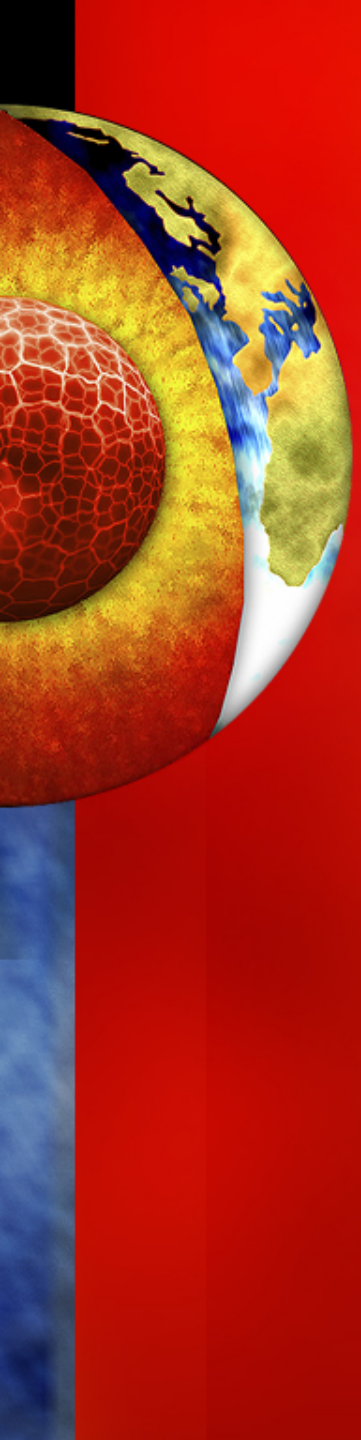


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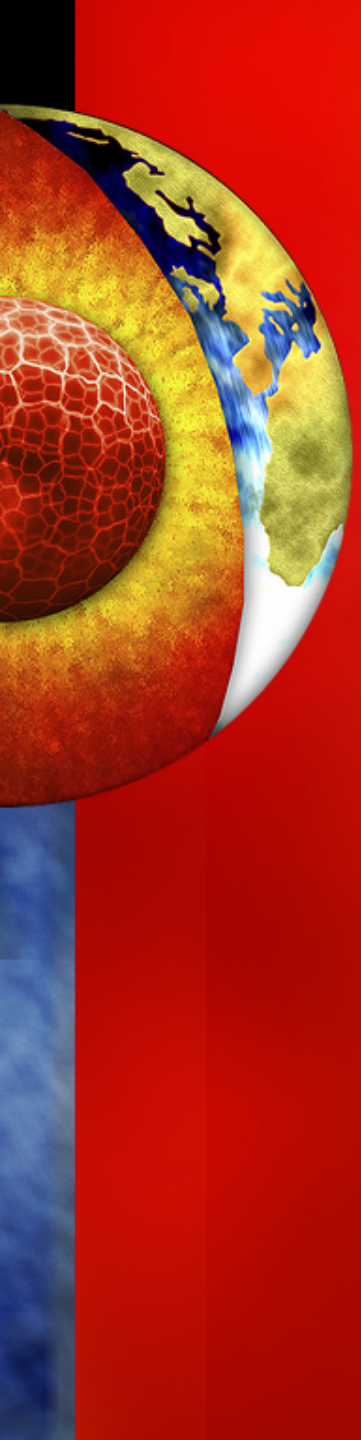


- Plate tectonics is a scientific theory that explains how major landforms are created as a result of Earth's subterranean movements. The theory, which solidified in the 1960s, transformed the earth sciences by explaining many phenomena, including mountain building events, volcanoes, and earthquakes.
- In plate tectonics, Earth's outermost layer, or lithosphere—made up of the crust and upper mantle—is broken into large rocky plates. These plates lie on top of a partially molten layer of rock called the asthenosphere.



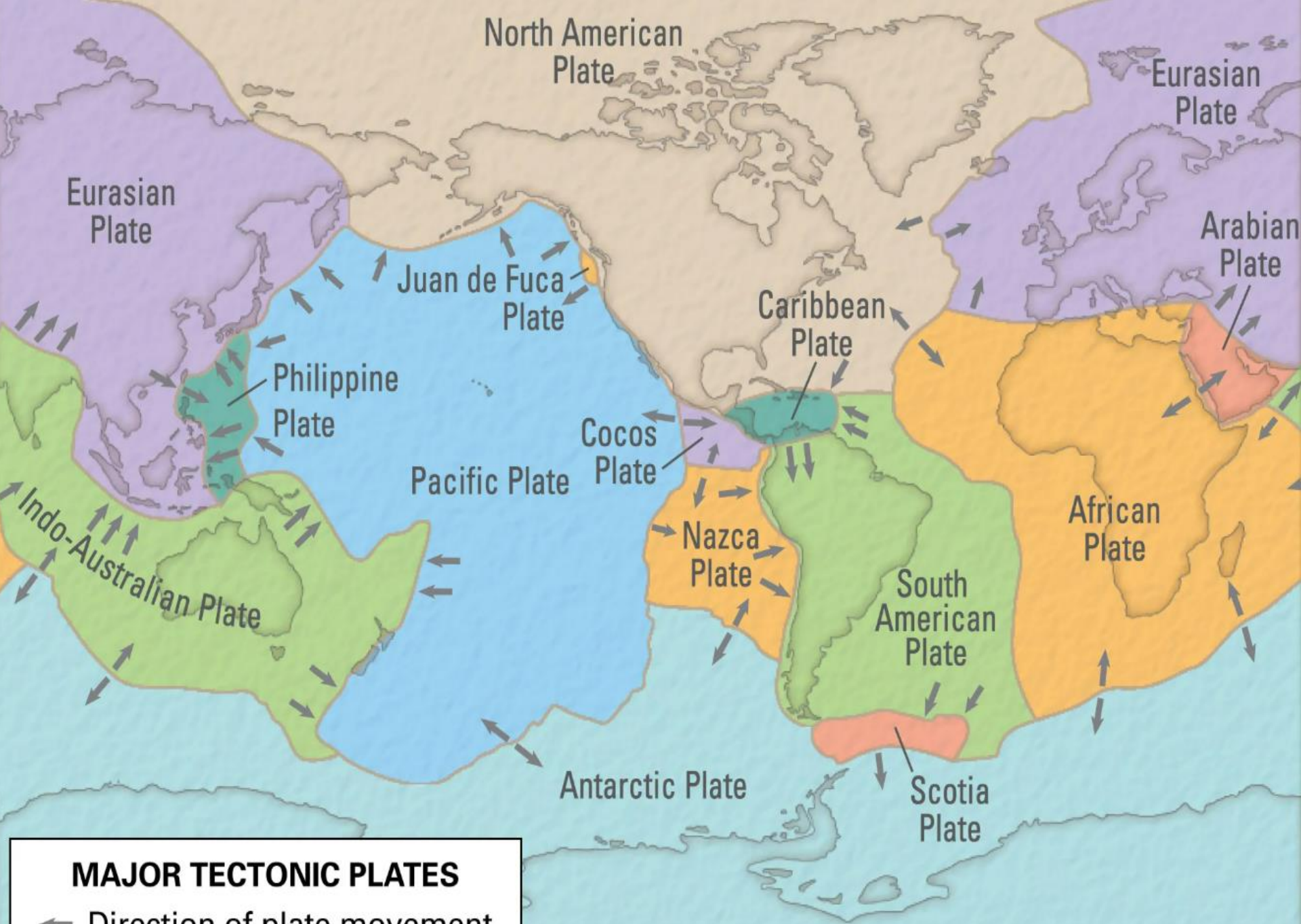
Large Tectonic Plates

- **Pacific Plate**
- **North American Plate**
- **Eurasian Plate**
- **African Plate**
- **Antarctic Plate**
- **Indo-Australian Plate**
- **South American Plate**



Small (or Minor) Tectonic Plates

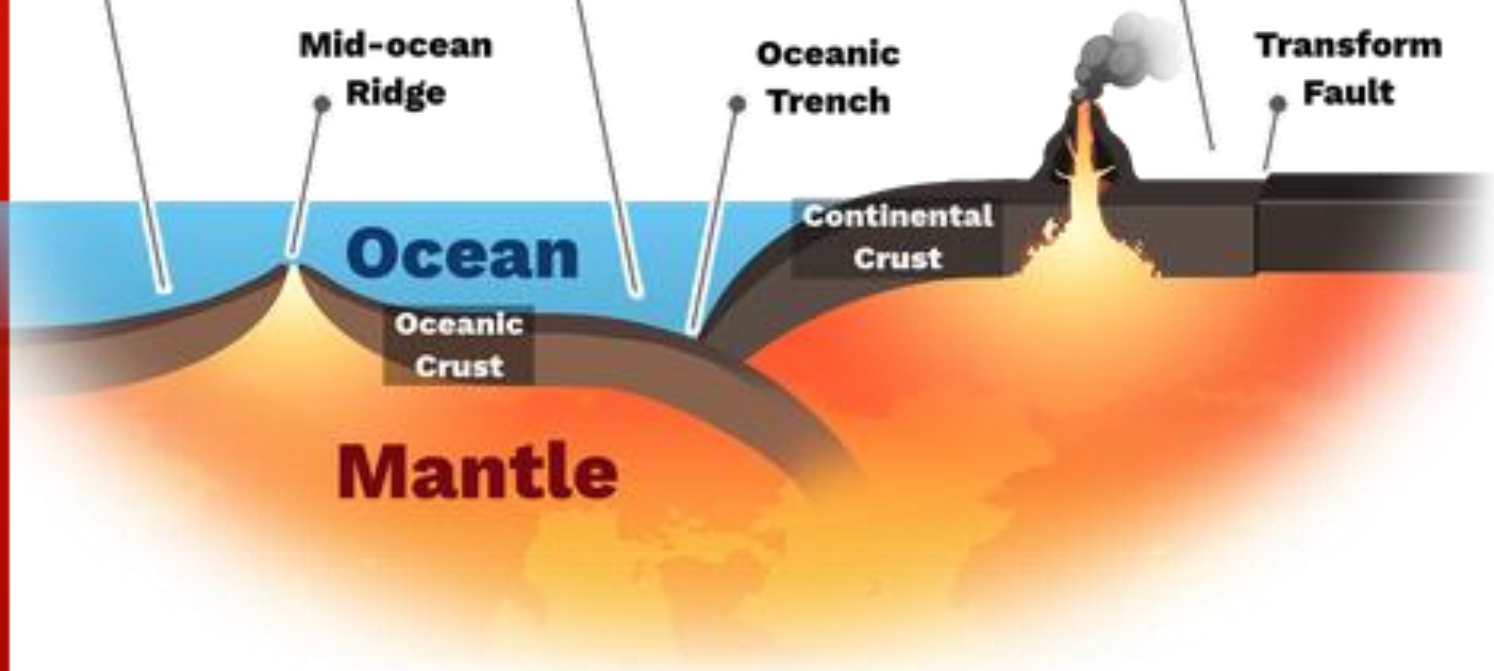
- Nazca Plate
- Cocos Plate
- Caribbean Plate
- Scotia Plate
- Philippine Sea Plate
- Arabian Plate
- Juan de Fuca Plate
- Caroline Plate
- Somali Plate
- Okhotsk Plate
- Burma Plate
- Anatolian Plate
- Balkan Plate
- Fiji Plate

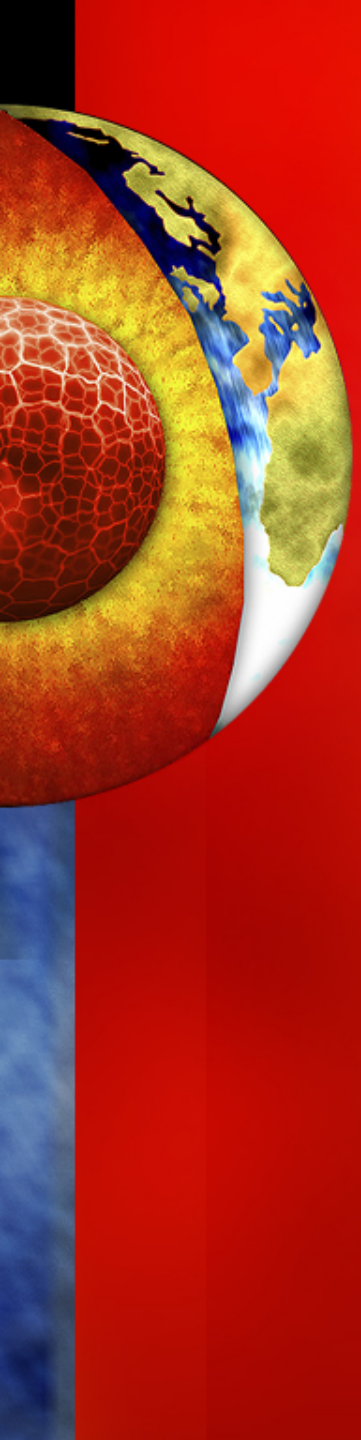


MAJOR TECTONIC PLATES

← Direction of plate movement

PLATE BOUNDARIES





Types of Plate Boundaries

Divergent boundaries

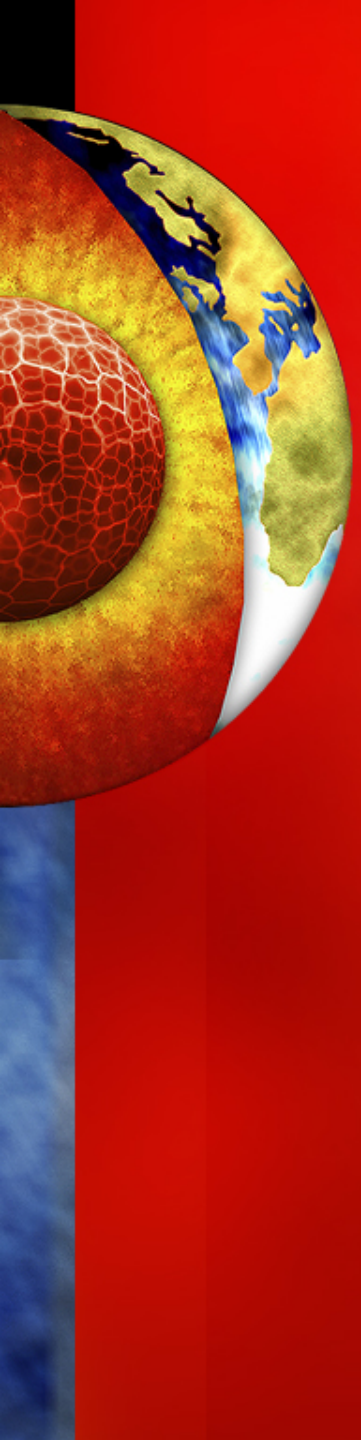
plates move apart, creating new crust (e.g., mid-ocean ridges).

Convergent boundaries

plates move toward each other, causing mountains or subduction (one plate sinks under another).

Transform boundaries

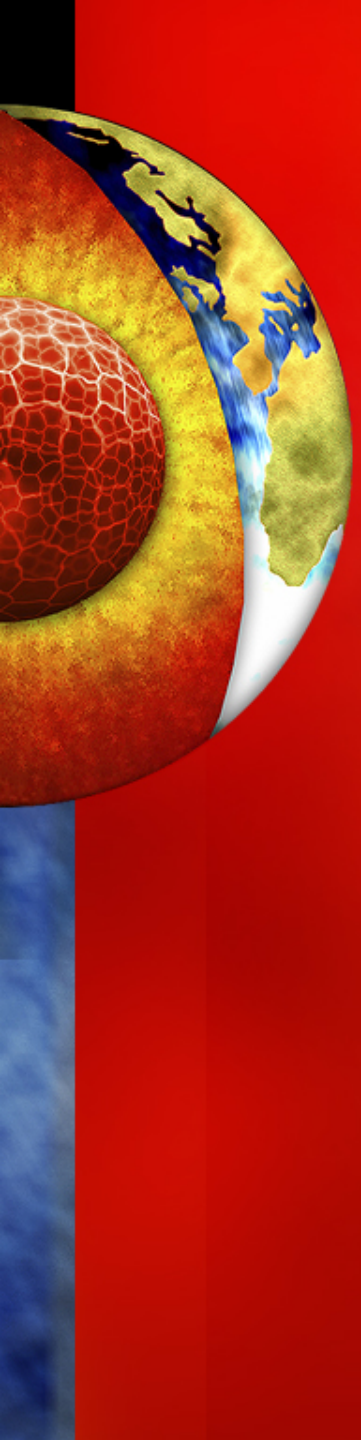
plates slide past each other, causing earthquakes (e.g., San Andreas Fault).



Driving Forces

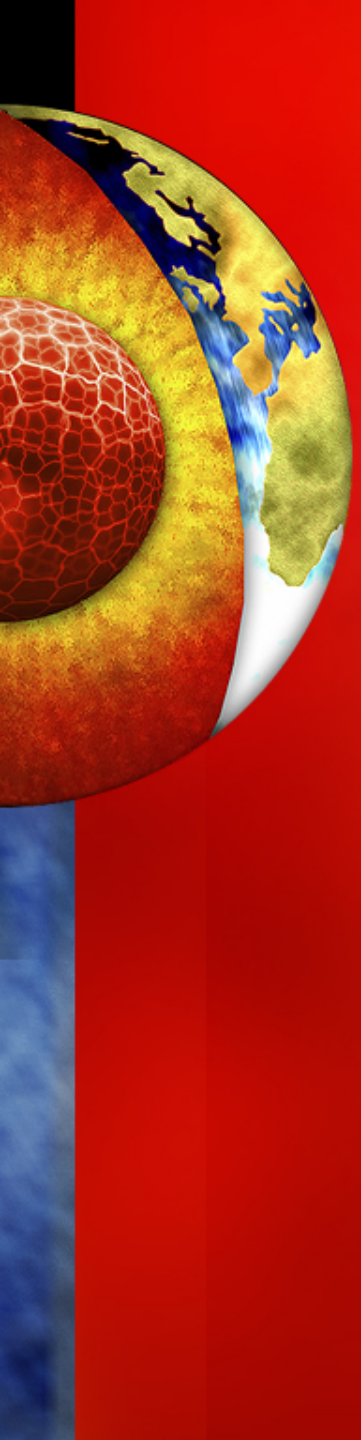
The movement of plates is powered by heat inside the Earth:

- Convection currents in the mantle (hot material rises, cools, and sinks).



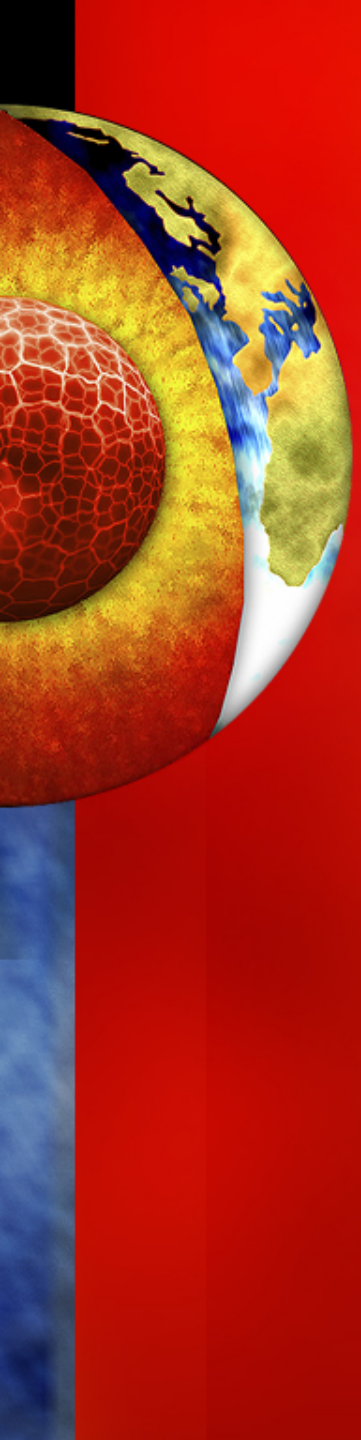
Effects of Plate Movements

- Creates mountains (like the Himalayas).
- Opens ocean basins and forms islands.
- Causes earthquakes and volcanic eruptions.
- Changes the positions of continents over millions of years (continental drift).



Evidence for Plate Tectonics

- Fit of continents (like South America and Africa).
- Fossils of the same species found on different continents.
- Patterns of earthquakes and volcanoes.
- Magnetic stripes on the ocean floor showing seafloor spreading.



Quiz

- Which layer of the Earth do the tectonic plates float on?

(Answer: Asthenosphere)

- What type of boundary creates mountains?

(Answer: Convergent boundary)

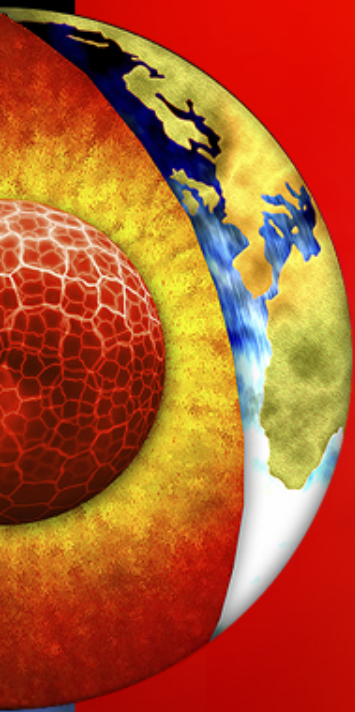
- Which boundary type is responsible for mid-ocean ridges?

(Answer: Divergent boundary)

- What is the name of the theory that explains the movement of continents?

-

(Answer: Plate Tectonics)



ANY
QUESTIONS



Thank
you!

